CycloBench: A Verifiable Benchmark for LLM Reasoning on Cyclostratigraphic Data

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Abstract

Large language models are increasingly used as scientific assistants, yet their ability to perform quantitative geoscientific analysis remains poorly measured. We introduce CycloBench, a verifiable benchmark for cyclostratigraphic analysis of sedimentary sequences. CycloBench pairs synthetic dual-well records with exact ground truth across six question families: sedimentation (duration and mean rate), age models, hiatus detection, orbital attribution of spectral peaks, amplitude modulation, and cross-well correlation. The benchmark contains 480 scored items from 60 instances (40 train / 20 test). Every item is scored 0/1 automatically, with two rules that block known shortcuts (the modern 41-kyr obliquity and single-proxy peak attribution). We release the generator, the instances, the scoring code and a fixed protocol with three conditions – zero-shot, chain-of-thought, and tool-augmented – designed to separate domain knowledge from tool use. Because the answers are verifiable, the benchmark also serves as a reward environment for post-training; we outline a GRPO recipe for a 1.5B model and release the code as a starting point for that use. In a first evaluation, five open models score between 0.0% and 15.6% macro accuracy, and neither chain-of-thought prompting nor a restricted Python sandbox closes the gap, so the benchmark remains far from saturation. Code and data are at https://github.com/KhanKehan/CycloBench.

1 Introduction

Large language models are increasingly used as scientific assistants, yet their ability to perform quantitative geoscientific analysis remains poorly measured. Existing benchmarks focus on textual knowledge or chart reading, not on the multi-step numerical workflow of cyclostratigraphy: spectral estimation with red-noise testing, sedimentation-rate modelling, astronomical tuning, and hiatus detection.

We argue that cyclostratigraphy is a useful testbed for verifiable scientific reasoning: data can be generated with known ground truth, answers are numeric or discrete, and correct solutions require both domain knowledge and tool use. This matters for evaluation because text-only benchmarks cannot distinguish a model that recites Milankovitch theory from one that can estimate a sedimentation rate from a log; it matters for post-training because a verifiable reward makes reinforcement learning possible without human preference labels, yet the reward must survive a multi-step analysis rather than a single answer string.

Our contributions are:

- CycloBench, a verifiable benchmark of 480 scored items across six task families, with exact ground truth, 0/1 scoring, and anti-cheat rules (§3).
- A fixed evaluation protocol with three prompting conditions and an error taxonomy that separates knowledge errors from tool-use and numerical errors (§4, §5).
- A verifiable-reward recipe for post-training, released as code, with the zero-shot evaluation as the baseline it must beat (§6).

2 Related Work

Benchmarks for language models. HELM [Liang et al., 2022] and BIG-bench [Srivastava et al., 2022] organise broad capability evaluations, while MMLU [Hendrycks et al., 2021] and GPQA [Rein et al., 2023]

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Family	Questions	Answer type	Tolerance
Sedimentation	Q1 duration, Q4 mean SR	float (Myr, cm/kyr)	0.10 / 0.20
Age model	Q2 age at 250 m	float (Ma)	0.050
Hiatus	Q3 depths and durations	list of (m, kyr)	3 m / 50 kyr
Orbital attribution	Q6 60/112-kyr peaks	4 booleans	exact
Modulation	Q7 amplitude modulation	bool + 2 floats	exact / range
Cross-well	Q8 shared hiatus	3 floats	3 m / 50 kyr

Table 1: Question families in CycloBench. Every instance asks all eight questions (Q1–Q8); Q1/Q4 share the sedimentation family and Q3/Q8 share the hiatus family.

test knowledge through multiple-choice questions. SciBench [Wang et al., 2023] moves to college-level problem solving but still scores a short final answer. None of these require the model to analyse raw measurements and return numbers that follow from the data, which is the setting of CycloBench. Agentic benchmarks such as ScienceAgentBench [Chen et al., 2024], MLE-bench [Chan et al., 2024], SciCode [Tian et al., 2024] and LAB-Bench [Laurent et al., 2024] evaluate end-to-end task completion or research coding; CycloBench instead isolates a single numerical workflow with exact, per-question ground truth.

Tool use and scientific workflows. ReAct [Yao et al., 2023] interleaves reasoning and actions, and Toolformer [Schick et al., 2023] teaches models to call APIs. Our tool-augmented condition follows this line, but the tool here is a numerical analysis stack over a data file, and the evaluation checks the final numbers rather than the trajectory. MassSpecGym [Bushuiev et al., 2024] is the closest benchmark in the adjacent mass-spectrometry domain and also centres on ranked structure candidates.

Cyclostratigraphy. Astronomical tuning relates sedimentary cycles to orbital forcing [Hinnov and Hilgen, 2007, Hinnov, 2013]. The 405-kyr eccentricity cycle provides a metronome, while obliquity and precession periods differ at 300 Ma from their modern values; hiatuses break phase continuity and must be detected before tuning. The standard toolchains, astrochron [Meyers, 2014] and Acycle [Li et al., 2019], implement these steps.

Post-training with verifiable rewards. GRPO [Shao et al., 2024] replaces the value network of PPO with group-normalised advantages and has become a standard recipe for reinforcement learning with verifiable rewards (RLVR) on math [Yang et al., 2024]. Process supervision [Lightman et al., 2023] and large post-training pipelines [Lambert et al., 2024] study how far such signals scale, while preference optimisation [Rafailov et al., 2023] and RLHF [Ouyang et al., 2022] address alignment rather than correctness. Whether RLVR transfers to scientific data analysis, where the reward requires running an analysis rather than matching a final answer string, is largely untested.

3 CycloBench

3.1 Task suite

CycloBench contains six task families (Table 1). Each item provides two wells of synthetic log data and asks for numeric or discrete answers that are checked exactly.

3.2 Data generation

Each instance is a forward simulation on a 1-kyr grid over $T \in [5.0, 7.8]$ Myr. Sedimentation rate follows a smooth low-frequency profile with a step change, drawn from 6.7–8.6 cm/kyr, and is set to zero during hiatus events (80–420 kyr each, constrained to be non-overlapping). Depth is the integral of the rate, so a hiatus produces a condensed interval in which age jumps while depth advances little.

Signals mix the 405-kyr and 100-kyr eccentricity terms, obliquity at the simulated 300-Ma period (32–36.5 kyr), and precession (18–21 kyr), with an AR(1) background. GR additionally carries autogenic 60- and 112-kyr cycles; CARB does not, and its obliquity amplitude is modulated at 170 kyr more strongly than GR. Well B shares the regional hiatus but has a different rate profile and its own local hiatuses, so the same event appears at different depths in the two wells.

Split	Instances	Duration (Myr)	Depth A (m)	SR (cm/kyr)	Hiatuses	Obliquity (kyr)
Train	40	5.0–7.8	291–597	5.5–7.8	3–5	32.0–36.4
Test	20	5.2–7.7	299–558	5.6–8.1	3–5	32.4–36.5

Table 2: Benchmark statistics. Each instance yields eight scored questions, so the benchmark contains 480 items in total.

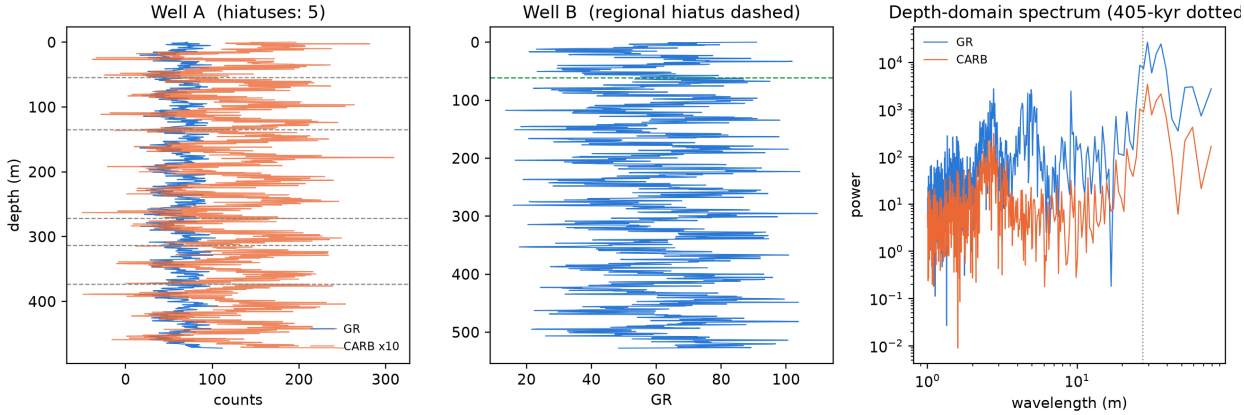


Figure 1: A CycloBenchinstance. Left: well A (GR and CARB) with the planted hiatus depths (dashed). Middle: well B, where the shared regional hiatus sits at a different depth. Right: depth-domain spectra; the dotted line marks the 405-kyr eccentricity wavelength implied by the mean sedimentation rate.

3.3 Scoring and anti-cheat

Each question is scored 0/1 with the tolerances in Table 1. Two rules target shortcuts: an answer of 41 kyr to Q5 is rejected because the obliquity period at 300 Ma differs from the modern value, and Q6 requires cross-validation between GR and CARB, so reporting a peak in one proxy alone does not earn credit. The final score of an instance is the fraction of its eight questions that pass. The generator writes the latent parameters to disk, so every truth value is known exactly rather than estimated, and generation is deterministic given the instance id (re-running the generator reproduces the same truth file byte for byte).

4 Experimental Setup

Models. We evaluate five open chat models spanning one order of magnitude in size: Qwen3-32B, DeepSeek-V3, GLM-4.6, Llama-3.1-70B, GPT-4o-mini, and a local Qwen2.5-1.5B-Instruct that also serves as the GRPO base. Qwen3-32B and Qwen3-14B were queried with hidden reasoning disabled (the vendor's enable_thinking=False); the other models ran with their default behaviour.

Conditions. Each instance is presented in one of three conditions. In zero-shot the model receives a data summary (size, depth range, sampling) and answers from knowledge. In chain-of-thought it is additionally asked to reason step by step. In tool the model receives the CSV paths and may execute Python (numpy/scipy/pandas) in a sandbox for up to three turns before returning the JSON.

Decoding and metrics. We use temperature 0 and a 2048-token limit, with four concurrent requests and three retries. The primary metric is the macro accuracy over the eight questions; we also report per-family accuracy (Table 3) and, for the tool condition, the fraction of instances where the executed code produced a finite result.

Training. The 40 training instances provide 320 question sets. GRPO uses a LoRA adapter on Qwen2.5-1.5B-Instruct with eight generations per prompt and two rewards: the fraction of correct answers (0–1) and a valid-JSON reward. We report benchmark accuracy on the 20 held-out instances plus GSM8K accuracy before and after training as a retention check.

Model	Zero-shot	CoT	Tool
Qwen3.5-122B	15.63	13.13	6.25
DeepSeek-V3.2	10.00	1.88	5.00
Qwen3-32B	0.00	1.88	2.50
GLM-4.5-Air	1.88	4.38	–
Qwen3-14B	2.50	1.25	–
Reference solver (non-LLM)	14.38	–	–

Table 3: Macro accuracy (%) over 20 held-out instances (160 scored questions per cell; parse failures count as zero; — marks a condition that did not complete). The reference solver is a Python implementation of the textbook workflow, not the generator.

Model (zero-shot)	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8
Qwen3.5-122B	0	5	0	0	80	35	5	0
DeepSeek-V3.2	0	0	0	0	35	45	0	0
Qwen3-14B	0	0	0	0	0	20	0	0
GLM-4.5-Air	0	0	0	0	0	10	5	0
Qwen3-32B	0	0	0	0	0	0	0	0

Table 4: Per-question pass rate (%) in the zero-shot condition. Q5 (the 300-Ma obliquity period) is the only numeric question solved at scale; Q6 (orbital attribution) is partially solvable; Q1–Q4 and Q8 are never solved.

Reproducibility. The generator is deterministic given the instance id: re-running it reproduces the truth file byte for byte. All result files are JSONL with the raw model output, the parsed answer and the per-question score; the analysis and tolerance-sweep scripts recompute every number in this section. API models were queried at temperature 0 with three retries; the tool sandbox preloads numpy, pandas and scipy and blocks file, network and import access.

5 Results

5.1 Main results

Table 3 reports the macro accuracy over the 20 held-out instances (160 scored questions per cell). No model comes close to solving the task: the best macro accuracy is 15.6% (Qwen3.5-122B, zero-shot), and three of the five models stay below 3%. A non-LLM reference implementation of the textbook workflow, written for this paper, reaches 14.4% on the same instances – the same range as the best model, which shows that the task is hard for careful hand-written pipelines as well as for LLMs. Chain-of-thought prompting does not help consistently (Qwen3.5-122B 15.6→13.1, DeepSeek 10.0→1.9, GLM-4.5-Air 1.9→4.4), and the tool condition is no better than zero-shot (best 6.3%). We return to why in §5.2. Bootstrap 95% confidence intervals over instances overlap for the top two models (Qwen3.5-122B 15.6 [11.9, 20.0], DeepSeek-V3.2 10.0 [6.9, 13.1]), so we do not claim a ranking between them; the gap to the remaining models is larger than its interval.

5.2 Error analysis

Across 260 model-instance runs, 94.6% of the outputs parse as JSON, so parsing is not what fails. The failures are numerical and structural. Q1–Q4 and Q8 almost never pass: models report durations an order of magnitude too large (e.g., 97.7 Myr against a 7.0-Myr record) and mean rates that are not on the right scale. Q5 (the 300-Ma obliquity period) is the only numeric question solved at scale: Qwen3.5-122B passes 80% of instances zero-shot, because the answer lies in a narrow, well-known band. Q6 is partially solvable (DeepSeek 45%), and Q7 is a binary direction that is occasionally guessed. Q1–Q4 and Q8 are zero for every model. The 41-kyr shortcut is rare (6 of 246 parsed answers), because most models do not answer with a plausible period at all. On Q3, 83.3% of answers report the wrong number of hiatuses, even when

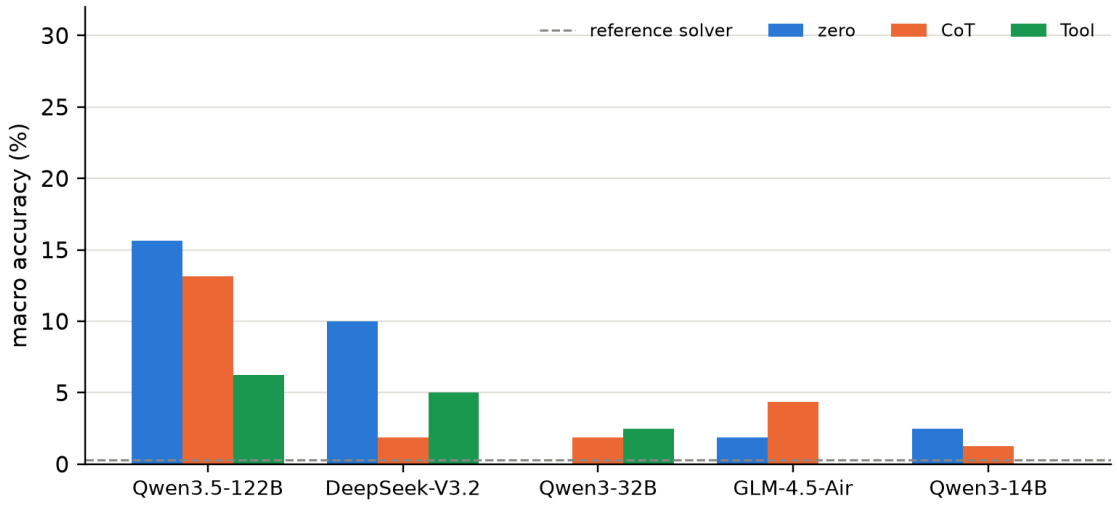


Figure 2: Macro accuracy by model and condition (20 held-out instances). The dashed line is the non-LLM reference solver; no model reaches half of it.

Question	Truth	Model answer	Condition
Q1 total duration	5.27 Myr	300.0 Myr	Qwen3-32B, tool
Q5 obliquity period	32.4 kyr	38.5 kyr	DeepSeek-V3.2, tool
Q3 hiatus count	3	2	DeepSeek-V3.2, CoT

Table 5: Representative errors. The first is a scale error of nearly two orders of magnitude; the second is a plausible but out-of-band period; the third is a list-level mistake.

individual depths are within a few metres; on Q6, 61.0% get one to three of the four booleans right, i.e., partial cross-validation rather than a coherent judgement. The tool condition does not fix this: the models run code, but the outputs are not turned into a tuned age model (best macro 6.3%). The analysis script is public and recomputes all of these counts from a result file.

To check whether the near-zero scores are an artefact of the tight tolerances, we re-scored the same answers with every tolerance multiplied by 2, 5 and 10. The gain is steady but small: Qwen3.5-122B rises from 15.6% to 23.1%, DeepSeek-V3.2 from 10.0% to 25.0%, and Qwen3-32B from 0.0% to 0.6%. The answers are far from the truth, not merely outside a narrow band.

6 Future Work: Verifiable-Reward Post-Training

Recipe. A natural use of CycloBench is as a reward environment. We fine-tune Qwen2.5-1.5B-Instruct with a LoRA adapter (r=16, alpha=32, all projection matrices) and train with GRPO on the 40 training instances (320 prompts). Each prompt asks for the full JSON answer, so one completion receives a reward equal to the fraction of its eight answers that the scorer accepts; a second reward gives partial credit for a parseable JSON object. We use eight generations per prompt, a completion limit of 768 tokens, $\beta = 0.04$, and a learning rate of 1×10^{-6} for one epoch. Training runs on a single A10 (23 GB) and is compared against the unadapted base on the 20 held-out instances.

6.1 Status

The recipe is released as code (train_grpo_cyclobench.py) with the same rewards and splits. We deliberately do not report trained-model numbers in this version: the evaluation in §5 is the baseline a training study must beat, and running it is the first item of future work.

7 Case Study: When the Tool Is Wrong

Tool-augmented evaluation assumes the tools are correct. In an earlier version of the benchmark we found a Bayesian changepoint function in an open-source cyclostratigraphy package that returned an all-zero posterior because the log-sum-exp was computed without the max-shift trick; the underflow made every candidate changepoint equally unlikely. Models that called the function and trusted it reported no hiatus, while models that inspected the posterior noticed the degenerate output. The bug was fixed by a numerically stable implementation, and synthetic tests then recovered all planted changepoints.

This episode motivates a design choice: the benchmark scores answers, not tool calls, so a model can still succeed by checking tool output. It also suggests an adversarial variant in which one tool is silently wrong; we leave that variant to future work and report only the correct-tool setting here.

8 Limitations and Conclusion

Our benchmark is synthetic, single-domain, and English-only. The generator imposes a known orbital model, so a model could in principle learn the generator's quirks rather than the science; the held-out instances share the generator but not the parameters, and the truth is computed from latent variables rather than from a solver, which limits circularity but does not remove it. Model coverage is a snapshot of the current API landscape.

A second limitation is calibration. The tolerances were set from the generator's exact parameters (e.g., 0.10 Myr on a 5–8 Myr record, 3 m on hiatus depth). A legitimate workflow that estimates the sedimentation rate locally can miss these bands even when its analysis is sound: our reference solver passes Q6 and Q7 almost always but never meets the numeric tolerances on Q1–Q4. The current benchmark therefore measures exact recovery, not approximate agreement, and part of the near-zero model scores reflects that choice. Two further caveats: Q5 (the obliquity period) can be answered from prior knowledge of the 300-Ma value, so a pass there is not evidence of data analysis; and the tool condition uses a restricted sandbox with three turns and a truncated output, which may understate what a better instrumented agent could do. Partial-credit variants and tolerance sweeps are the obvious next step.

Within these limits, CycloBench shows that verifiable scientific workflows can be measured with exact answers, that current models score near zero without tools, and that a restricted sandbox does not close the gap by itself. We release the generator, the 60 instances, the scoring code, and the training recipe.

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